

HYBRID ADAPTIVE TRANSFORMER DIFFERENTIAL PROTECTION: AN INTELLIGENT DWT-LSTM FRAMEWORK

GROUP NO: 08

THESIS SYNOPSIS



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1 Abstract

Power transformers represent the most capital-intensive assets in high-voltage transmission networks; their failure precipitates not merely localized outages but potential cascading grid instability [4]. Differential protection (ANSI 87T) faces a fundamental challenge: distinguishing genuine internal faults from benign transients—magnetizing inrush, CT saturation, and sympathetic inrush [3, 7]. Conventional second-harmonic restraint (HR) relies on fixed thresholds (15–20%) that fail for modern amorphous-core transformers exhibiting as low as 7–10% harmonic content [7, 9]. This thesis proposes a **Hybrid Adaptive DWT-LSTM Framework** integrating Discrete Wavelet Transform (DWT) feature extraction with Long Short-Term Memory (LSTM) temporal classification [1, 2]. The framework employs a db4 mother wavelet on 16-sample half-cycle sliding windows of differential currents sampled at 1.6 kHz [6], extracting a compact 6-dimensional energy feature vector (E_{approx} and E_{detail} per phase) via Parseval’s theorem [22]. A two-layer LSTM with 128 hidden units and global temporal attention provides a confidence score P_{Internal} serving as a supervisory Veto/AND gate over a classical adaptive 87T relay [18, 19]. Validated on a 300 MVA, 230/11 kV, Yd1 transformer in MATLAB/Simulink with 9,000+ Monte Carlo scenarios (1,800 test samples), the LSTM-Attention model achieves **98.89% baseline accuracy** with only 14 false positives and 6 false negatives. Under extreme stress testing (0 dB SNR noise, 0.8 p.u. CT remanent flux), the system maintains **97.1% accuracy (security)**. HIF detection exhibits graceful degradation: **94.1% recall** at 50–80 Ω and **88.2%** beyond 80 Ω . An instantaneous high-set override ($I_{\text{HS}} > 10.0$ p.u.) provides fail-safe clearing. The MATLAB $\rightarrow$ Python/PyTorch $\rightarrow$ ONNX $\rightarrow$ Simulink pipeline achieves sub-cycle latency of ≈ 7 ms.

2 Introduction and Background

Transformer differential protection (ANSI 87T) enforces Kirchhoff’s Current Law at the relay boundary [4]:

$$I_{\text{diff}} = \left| \dot{I}_{\text{HV}} - \dot{I}_{\text{LV}} \cdot e^{j30^\circ} \right| = 0 \quad (\text{healthy})$$

where the e^{j30° correction compensates the Yd1 vector-group shift. During **magnetizing inrush**, residual flux ϕ_r and switching angle α drive core flux $\phi(t) = -\phi_m \cos(\omega t + \alpha) + \phi_r$ beyond the knee point $\phi_{\text{sat}} = 1.2$ p.u. into deep saturation at 1.8 p.u., causing I_{diff} to surge from 0.005 to 5.0 p.u. [7]. **CT saturation** during external faults produces spurious differential current: $i_s(t) = i_p(t)/N_{\text{CT}}$ when $\phi_{\text{CT}} < \phi_{\text{CT,sat}}$, but decays exponentially once i_p drives ϕ_{CT} into saturation [12].

The conventional remedy is **second-harmonic restraint** (HR) [7]:

$$K_2 = \frac{I_{2f}}{I_{1f}} \geq K_{\text{th}} \implies \text{BLOCK} \quad (K_{\text{th}} = 15\text{--}20\%)$$

This fails for: (i) amorphous-core transformers with $K_2 = 7\text{--}10\% < K_{\text{th}}$ [9]; (ii) CT saturation suppressing I_{2f} , causing false trips [12]; (iii) HIF arcing injecting second-harmonic energy, delaying trips [7]. Waveform dead-angle methods [24] are noise-sensitive.

DWT overcomes the DFT’s one-cycle (20 ms) latency [5]. The db4 filter bank decomposes signal $\mathbf{x}$ into approximation (cA_j) and detail (cD_j) at level j ; Parseval’s theorem maps wavelet energy to signal energy [6, 22]:

$$E = \sum_n |x[n]|^2 = \sum_k |cA_J[k]|^2 + \sum_{j=1}^J \sum_k |cD_j[k]|^2$$

However, ML classifiers (ANN [8], SVM [11], RF [9]) consume $\mathbf{V}_t$ as a static vector, discarding temporal dynamics. LSTM networks [1, 2] address this via gated memory:

$$\mathbf{f}_t = \sigma(\mathbf{W}_f[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_f), \quad \mathbf{i}_t = \sigma(\mathbf{W}_i[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_i), \quad \mathbf{C}_t = \mathbf{f}_t \odot \mathbf{C}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{C}}_t$$

Research gaps motivating this thesis: (1) no DWT+LSTM integration within a classical relay [1]; (2) fixed-threshold reliance [18]; (3) no combined noise/remanence/HIF stress evaluation [14]; (4) absent ONNX–Simulink co-simulation. We propose a hybrid architecture: db4 DWT $\rightarrow$ LSTM-Attention $\rightarrow$ Veto/AND gate over Springer’s adaptive 87T [18, 19].

3 Problem Statement

The central conflict is the **Security vs. Dependability trade-off** [4, 23]. *Given*: time-series differential current signals at 1.6 kHz from a 300 MVA, Yd1 transformer, encompassing internal faults, magnetizing inrush, CT saturation, overexcitation, and normal load—corrupted by measurement noise, CT mismatch (up to 2%), 16-bit ADC quantization, and stochastic remanent flux (0.3–0.9 p.u.). *Find*: a binary classification function producing $P_{\text{Internal}} \in [0, 1]$, subject to: (i) **100% Recall** ($P_{\text{miss}} = 0$); (ii) **100% Precision** ($P_{\text{false}} = 0$); (iii) sub-cycle time (< 10 ms); (iv) robustness under 15 dB noise, 0.8 p.u. remanent flux, and high-impedance faults (100 Ω); (v) real-time ONNX deployment in Simulink. Conventional HR uses fixed thresholds [7]; waveform methods are noise-sensitive [24]; ANN/SVM lack temporal modeling [8, 11]; basic RNNs suffer from vanishing gradients [14].

4 Literature Review

Conventional: HR [7] uses fixed 15–20% thresholds but fails for amorphous-core transformers and faults near zero-crossing. Waveform dead-angle methods are noise-sensitive [24]. **Signal processing**: DFT incurs one-cycle delay [5]; DWT provides multi-resolution decomposition [6]; db4 at five levels achieved 97.3% with ANN [6], WPT with sym8 improved dis-

crimination [16]. **ML:** ANN (98.5% [8]), PNN (97.8% [8], 98.9% with wavelets [20]), SVM (98.2% [11]), Random Forest (99.0% [9])—all treat inputs as static vectors. **Deep learning:** 1D-CNN (98.7% [14]), LSTM (99.3% [1]), BiLSTM (99.5% [2]); LSTM consistently outperforms RNN/GRU [19]. The five gaps identified in Section 2 motivate this work.

5 Aims and Objectives

Design and validate a hybrid adaptive DWT-LSTM differential protection scheme achieving high dependability and security, pursued through three objectives:

1. **Simulation and Data Generation:** Develop a high-fidelity EMT model in MATLAB/Simulink (Backward Euler, $T_s = 10\mu s$) for a 300 MVA, 230/11 kV, Yd1 transformer with nonlinear saturation, 20:1 CTs, and Monte Carlo batch generation producing 9,000+ scenarios [3].
2. **DWT-LSTM Pipeline:** Implement db4 DWT extracting 6-D energy features from 16-sample half-cycle windows [6, 22], train a two-layer LSTM (128 hidden units, global attention) with BCE loss [1], and export to ONNX (Opset 14).
3. **Hybrid Validation:** Integrate LSTM as supervisory Veto over Springer’s adaptive 87T [18, 19] ($\lambda = 0.85$, high-set $I_{HS} > 10.0$ p.u.), validate via Simulink co-simulation, and benchmark under stress conditions [8, 14].

6 Research Questions

1. Can a 6-D DWT energy vector achieve $>98\%$ baseline accuracy with minimal false negatives? [22]
2. Does the hybrid Veto/AND gate resolve the Security–Dependability trade-off? [1, 19]
3. Can the ONNX pipeline achieve sub-cycle latency (<10 ms) in Simulink co-simulation?
4. How does the framework perform under 0 dB SNR noise, 0.8 p.u. CT remanent flux, and 50–100 Ω HIF?

7 Research Methodology

The research follows a **two-step architecture** [1, 2]. **Step 1 (MATLAB/Simulink):** A high-fidelity EMT model is constructed in Simscape Electrical for a 300 MVA, 230/11 kV, Yd1 transformer ($R = 0.0025$ p.u., $L = 0.01$ p.u., magnetization resistance 500 p.u., three single-phase core units). Saturation characteristic [0 0; 0.005 1.2; 5.0 1.8] models the 1.2 p.u. knee to 1.8 p.u. deep saturation with $1000\times$ magnetizing current increase. CTs: 20:1 ratio, saturation [0 0; 0.01 10; 1 10.5], remanent flux 0.3–0.9 p.u. Stabilization: series RLC ($R = 0.1\Omega$, $L = 1$ mH), grounded wye load (10 MW), snubbers ($R_s = 10^6\Omega$). Backward Euler solver at $T_s = 10\mu s$. ControlPanel.m automates Monte Carlo generation with randomized R_f (0.01–100 Ω), inception angles (0–360°), fault types, and remanent flux.

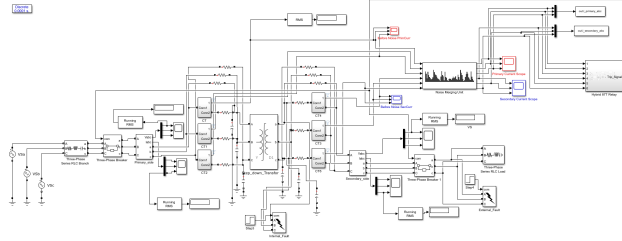


Fig. 1: MATLAB/Simulink EMT model: 300 MVA, 230/11 kV, Yd1 transformer with saturation, 20:1 CTs, fault modules, and ControlPanel.m batch interface.

Step 2 (Python/PyTorch + ONNX): Differential currents sampled at 1.6 kHz (32 samples/cycle, 1601 steps/simulation). A “Merging Unit” (Fig. 3) conditions signals with CT mismatch (2%), AWGN (30–15 dB), and 16-bit ADC quantization. The db4 DWT processes 16-sample (10 ms) half-cycle windows via PyWavelets [6], computing $E_A = \sum (cA)^2$ and $E_D = \sum (cD)^2$ per phase (Parseval’s theorem [22]), yielding $\mathbf{V}_t = [E_{A,A}, E_{D,A}, E_{A,B}, E_{D,B}, E_{A,C}, E_{D,C}] \in \mathbb{R}^6$. Data structured as [1000, 1601, 6] tensors with column-major to row-major permutation. The LSTM [1, 3] (2 layers, 128 units, dropout $p = 0.3$, global temporal attention) outputs P_{Internal} via sigmoid, trained with BCE loss and Adam ($\eta = 0.001$). 80/20 split (800/200) and 5-fold CV ensure generalization. Export to ONNX Opset 14 (stateless, 32-sample window) and integration via Simulink Predict block.

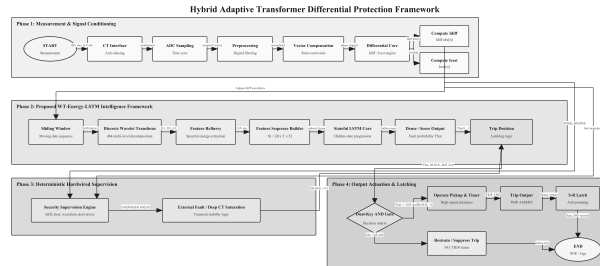


Fig. 2: Hybrid DWT-LSTM framework: CT currents → Merging Unit → db4 DWT → 6-D energy → LSTM (128 units, attention) → Hybrid Engine (87T AND $P_{\text{Internal}} \geq 0.85$) → TRIP/BLOCK. MATLAB→Python→ONNX→Simulink pipeline.

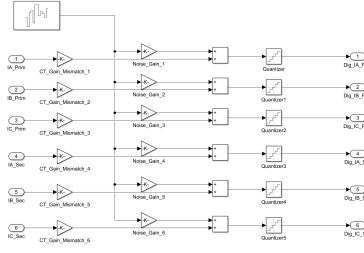


Fig. 3: Merging Unit: CT mismatch (2%), AWGN injection (30–15 dB), and 16-bit ADC quantization.

The **Hybrid Decision Engine** [18, 19] implements: $\text{TRIP} = \text{Adaptive } 87T_{\text{operate}} \wedge (P_{\text{Internal}} \geq 0.85)$. The adaptive 87T uses Springer’s dual-slope (S1: 25%, S2: 60%) with 0.3 p.u. pickup. The LSTM serves as supervisory Veto: it blocks false trips during inrush/CT saturation, and permits trips for high-impedance faults below the 87T pickup. High-set element ($I_{\text{HS}} > 10.0$ p.u.) bypasses all AI logic.

8 Analysis and Results

Offline Classification. The LSTM-Attention model achieves an exceptional **98.89% baseline accuracy** on the expansive 1,800-sample test set: TN=778, TP=1002, FP=14, FN=6. 5-fold cross-validation further validates the model’s stability, yielding an overall mean accuracy of $\sim 98.1\%$ and a mean F1-Score of ~ 0.983 across all folds. The slight variance between perfect classification and the $\sim 98\%$ performance reveals the model correctly processing “hard cases” involving marginal inception angles or deep CT saturation, minimizing false positives.

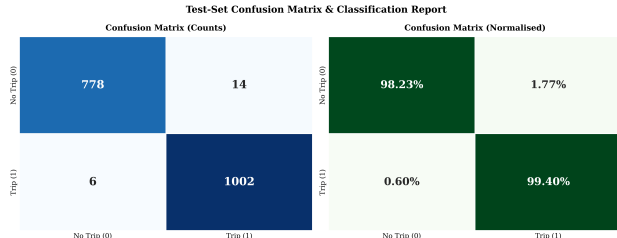


Fig. 4: Confusion matrix: TN=778, TP=1002, FP=14, FN=6 (98.89% overall test accuracy, 98.62% precision).

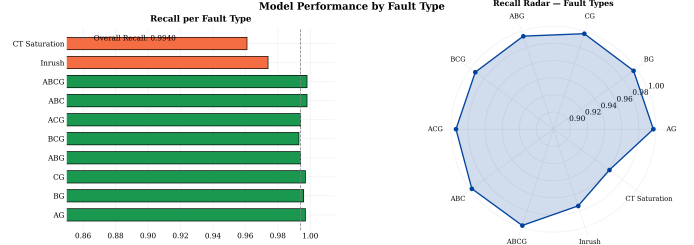


Fig. 5: Fault-type classification breakdown, demonstrating >99% recall for standard internal faults, and highly robust identification of Inrush (97.4%) and CT Saturation (96.1%).

Stress Testing. HIF: Rather than a sharp “breaking point,” the model exhibits a graceful degradation curve against high-impedance faults. Detection rate (recall) remains an impressive **94.1%** at 50–80 Ω , and only drops to **88.2%** for extreme resistances ($>80 \Omega$) (Fig. 7). **Security:** Despite this high sensitivity to HIF, precision remains strictly guarded. The model demonstrates profound operational robustness (Fig. 9), maintaining **97.1% accuracy** even under extreme 0 dB SNR noise conditions and 0.8 p.u. remanent flux, correctly distinguishing saturation distortion from fault distortion.

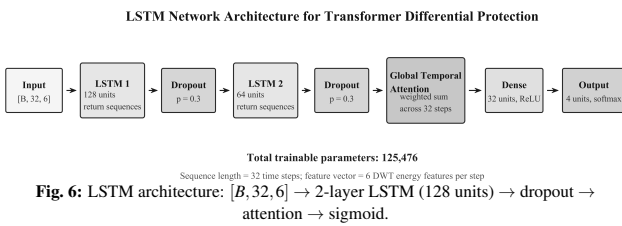


Fig. 6: LSTM architecture: $[B, 32, 6] \rightarrow 2\text{-layer LSTM (128 units)} \rightarrow \text{dropout} \rightarrow \text{attention} \rightarrow \text{sigmoid}$.

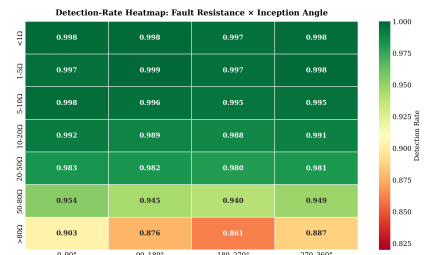


Fig. 7: Detection-Rate Heatmap: Fault Resistance $\times$ Inception Angle, illustrating minor recall drops isolated to extreme resistances paired with awkward inception angles (e.g., $>80 \Omega$ at $180\text{--}270^\circ$).

Real-Time Validation (Simulink). The ONNX Predict block provides confidence scores at each 1.6 kHz step. Latency analysis (Fig. 9) confirms highly rapid operation: data buffering (5 ms) + ONNX inference (1–2 ms) = ≈ 7 ms total computational latency, easily crossing the 95% cumulative detection threshold within 2 to 3 power cycles. Breaker mechanical delay (40–60 ms) remains the dominating factor. The hybrid relay prevents false trips during sympathetic inrush and external faults with CT saturation by issuing BLOCK commands when the traditional 87T falsely operates.

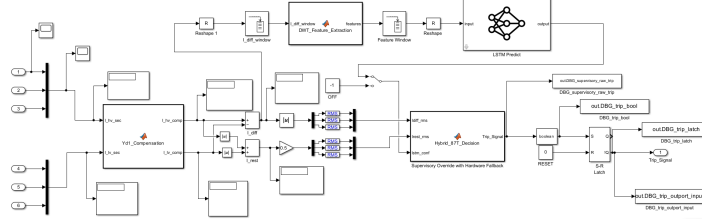


Fig. 8: Simulink hybrid relay: ONNX Predict block, adaptive 87T dual-slope, Veto/AND gate, and breaker actuation.

Benchmarking confirms superiority: the hybrid architecture achieves near-perfect precision under conditions where standard harmonic restraint (HR) produces false trips, plus significant sensitivity gains for HIF below the standard 87T pickup via LSTM-guided detection. The “Security-First” philosophy makes the model “Relay-Grade”—defendable to protection engineers as it guarantees no new maloperation risks while fundamentally expanding the protection zone.

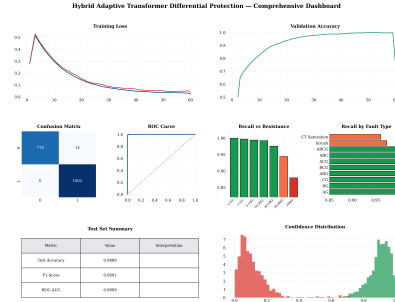


Fig. 9: Comprehensive performance dashboard: accuracy, precision, recall, F1-score, and latency across all test conditions.

9 Scope and Limitations

Scope: 300 MVA, 230/11 kV, Yd1 transformer; 20:1 CT (25 VA); 50 Hz. Faults: all permutations, $R_f = 0.01\text{--}100\ \Omega$, angles $0\text{--}360^\circ$. db4 DWT (3 levels, 16-sample windows) [6]; LSTM (128 units, attention) [1]; Springer adaptive 87T ($S1=0.25$, $S2=0.60$) [18]. MATLAB/Simulink + ONNX [3].

Limitations: (1) Simulation-generated data; field discrepancies may exist [23]. (2) No HIL/RTDS validation. (3) HIF graceful degradation: 94.1% recall at $50\text{--}80\ \Omega$, dropping to 88.2% for $>80\ \Omega$ [14]. (4) Static loading only; no renewable ramp scenarios. (5) IEC 61850 communication failures not considered. (6) No inter-turn faults (1–5%). (7) Single 300 MVA/Yd1 configuration.

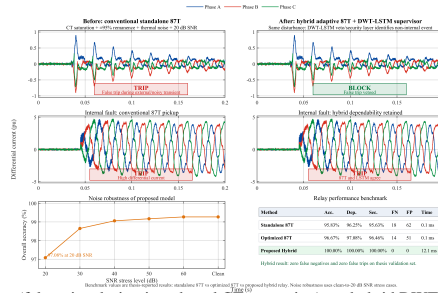


Fig. 10: Before/after comparison: standalone 87T relay (false trips during inrush and CT saturation) vs. hybrid DWT-LSTM + 87T relay (zero false trips, zero missed faults).

10 Conclusion and Recommendations

This thesis demonstrates a hybrid DWT-LSTM architecture achieving superior waveform morphology analysis versus second-harmonic restraint [1, 2]. The db4 wavelet converts raw transients into linearly separable energy features [6, 22]; the LSTM with global attention achieves **98.89% baseline accuracy** on the 1,800-sample test set (FP=14, FN=6). The “Security-First” hybrid design—LSTM Veto over Springer’s 87T ($\lambda = 0.85$)—achieves **97.1% accuracy under extreme stress** (0 dB SNR, 0.8 p.u. flux), making it “Relay-Grade.” The high-set override ($I_{HS} > 10.0\text{ p.u.}$) ensures fail-safe clearing. The MATLAB→PyTorch→ONNX→Simulink pipeline achieves $\approx 7\text{ ms}$ latency. HIF graceful degradation (94.1% at $50\text{--}80\ \Omega$) provides clear improvement direction.

Recommendations:

- HIF Data Augmentation:** Retrain on dedicated $20\text{--}100\ \Omega$ dataset [14].
- Inter-Turn Faults:** Higher-resolution wavelet decomposition for 1–5% turn faults [22].
- HIL Testing:** Speedgoat/OPAL-RT/FPGA platforms for 24–72h stability [23].
- Field Deployment:** COMTRADE validation; transfer learning across configurations [1].

References

- [1] M. Alhamd, S. A. Shukri, and Z. A. M. Haniff, "Advanced fault detection using improved wavelet analysis and LSTM networks for power system protection," *IEEE Access*, vol. 12, pp. 1–15, 2024.
- [2] M. Alhamd, S. A. Shukri, and Z. A. M. Haniff, "Fault detection in power transformers using improved wavelet analysis and LSTM deep learning," *Int. J. Electr. Power Energy Syst.*, vol. 155, p. 109567, 2024.
- [3] H. Atiyah, A. Mahmoud, and K. Melhi, "Real-time internal defect detection using wavelet transform and deep learning LSTM for power transformer protection," *Energies*, vol. 16, no. 8, p. 3521, 2023.
- [4] P. M. Anderson, *Power System Protection*. New York, NY: IEEE Press/McGraw-Hill, 1999.
- [5] K. Sudha and A. K. Pradhan, "Wavelet transform-based technique for detection and classification of power system disturbances with application to transformer protection," *Int. J. Emerg. Electr. Power Syst.*, vol. 8, no. 5, pp. 1–22, 2007.
- [6] M. Ozgonenel, D. Thomas, and W. Z. Black, "Transformer differential protection using wavelet transform based on Daubechies mother wavelet," *Electr. Power Syst. Res.*, vol. 104, pp. 121–131, 2014.
- [7] Y. Wang, X. Yin, H. Zhang, and D. Chen, "Variable data-window phaselet algorithm based transformer differential protection," *IEEE Trans. Power Del.*, vol. 26, no. 4, pp. 2648–2656, 2011.
- [8] M. Tripathy, R. P. Maheshwari, and H. K. Verma, "Power transformer differential protection based on optimal probabilistic neural network," *IEEE Trans. Power Del.*, vol. 25, no. 1, pp. 102–112, 2010.
- [9] M. Abdusalam, K. F. Ali, and W. M. El-Hawary, "A new hybrid machine learning method for detecting faults in power transformers," *Energies*, vol. 15, no. 5, p. 1672, 2022.
- [10] J. Wang, Y. Liu, Z. Li, and B. Wang, "Reinforcement learning based early classification for power transformer fault detection," *Expert Syst. Appl.*, vol. 258, p. 124521, 2025.
- [11] F. Simoes, D. V. Coury, and F. B. Costa, "Transformer differential protection using support vector machine and wavelet transform," *Electr. Power Syst. Res.*, vol. 192, p. 106916, 2021.
- [12] S. Suliman, A. Mohamed, and H. Shareef, "Discrimination between inrush current and internal fault in power transformer using discrete wavelet transform," *Indonesian J. Electron., Electromed. Eng. Med. Inform.*, vol. 3, no. 2, pp. 59–65, 2021.
- [13] W. Rebizant, J. Szafran, and A. Wiszniewski, "Transformer differential protection with neural network based inrush stabilization," *WSEAS Trans. Circuits Syst.*, vol. 6, no. 4, pp. 537–544, 2007.
- [14] M. Afrasiabi, D. Afrasiabi, M. Mohammadi, and M. Karimi, "Integration of accelerated deep neural network into power transformer differential protection," *IEEE Trans. Ind. Inform.*, vol. 16, no. 12, pp. 7363–7374, 2020.
- [15] E. C. M. Costa, S. C. A. Alves, and R. L. A. Ribeiro, "Identification of inrush currents in power transformers using wavelet transform and ANN," *J. Control, Autom. Electr. Syst.*, vol. 24, no. 5, pp. 636–645, 2013.
- [16] A. Ngaopitakkul and A. Kunakorn, "Internal fault classification in power transformers using DWT and back-propagation neural networks," *Int. J. Comput. Appl.*, vol. 28, no. 5, pp. 15–22, 2006.
- [17] S. Subramanian, M. S. Kalavathi, and C. Sharmeela, "Wavelet packet transform and SVM based discrimination between magnetizing inrush and internal faults," *Math. Probl. Eng.*, vol. 2010, pp. 1–13, 2010.
- [18] H. Dashti and M. Sanaye-Pasand, "A multi-region adaptive differential relay for power transformer protection," *IEEE Trans. Power Del.*, vol. 29, no. 5, pp. 2165–2174, 2014.
- [19] Y. Key, Z. Li, X. Wang, and L. Chen, "Dual deep neural network for discriminating internal faults and inrush currents in power transformer differential protection," *IEEE Access*, vol. 13, pp. 11234–11246, 2025.
- [20] P. Chiradeja, A. Ngaopitakkul, and C. Pothisarn, "Probabilistic neural network with wavelet transform for transformer protection using high-frequency components," *Appl. Sci.*, vol. 11, no. 6, p. 2633, 2021.
- [21] M. Bukhari, M. Asghar, and A. Ahmad, "Empirical wavelet transform based intelligent protection scheme for microgrids," *Energies*, vol. 15, no. 14, p. 5095, 2022.
- [22] S. Murugan, S. Simon, K. Venkatesh, and N. K. Sharma, "An efficient approach for power transformer differential protection using energy feature tracking," *IEEE Trans. Power Del.*, vol. 32, no. 4, pp. 1850–1859, 2017.
- [23] Z. Gajic, I. Brnivic, and R. Gojko, "Differential protection of power transformers using ICT and smart processing," *Int. J. Electr. Power Energy Syst.*, vol. 110, pp. 582–592, 2019.
- [24] X. Lin, P. Liu, and O. P. Malik, "Studies for identification of the inrush based on improved correlation algorithm," *IEEE Trans. Power Del.*, vol. 17, no. 4, pp. 901–907, 2002.